Formulae, calculations and ionic bonding

25 questions: 8 multiple choice, 17 structured.

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1 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [1]

Ammonium sulfate, $(NH_4)_2SO_4$, is used as a fertiliser. Use the Periodic Table.

What is the relative formula mass (Mr) of ammonium sulfate?

A 114

B 132

C 128

D 228

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2 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [1]

Use the Periodic Table.

Which of these samples contains the greatest amount of substance, in moles?

A 25 g of calcium carbonate, $CaCO_3$

B 16 g of sulfur dioxide, SO_2

C 12 g of oxygen, O_2

D 9.0 g of water, H_2O

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3 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [1]

The molecular formula of a compound is $C_6H_{12}O_2$.

What is the empirical formula of this compound?

A C_3H_6O

B CH_2O

C $C_6H_{12}O_2$

D $C_3H_6O_2$

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4 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [1]

Hana needs to make 250 cm^3 of sodium hydroxide solution, NaOH(aq), with a concentration of 0.200 mol/dm^3. Use the Periodic Table.

What mass of sodium hydroxide does she need to dissolve?

A 8.00 g

B 32.0 g

C 2.00 g

D 2000 g

5 · Part 2

Paper 2 only | Chemical formulae, equations and calculations | [1]

Magnesium reacts with excess dilute hydrochloric acid.

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

In an experiment, 0.36 g of magnesium reacts completely. Use the Periodic Table. (1 mol of any gas occupies 24 dm^3 at rtp)

What volume of hydrogen, measured at rtp, is produced?

A 360 cm^3

B 180 cm^3

C 720 cm^3

D 8640 cm^3

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6 · Part 2

Paper 1 & 2 | Ionic bonding | [1]

What is the formula of aluminium sulfate?

A $AlSO_4$

B $Al_3(SO_4)_2$

C $Al_2(SO_4)_3$

D $Al(SO_4)_3$

7 · Part 2

Paper 1 & 2 | Ionic bonding | [1]

Magnesium is in Group 2 of the Periodic Table.

What happens when a magnesium atom forms a magnesium ion?

A. It gains two electrons and forms Mg^{2-}.

B. It loses two protons and forms Mg^{2+}.

C. It shares two electrons with another atom.

D. It loses two electrons and forms Mg^{2+}.

8 · Part 2

Paper 1 & 2 | Ionic bonding | [1]

Which of these describes an ionic bond?

A. the attraction between a shared pair of electrons and two nuclei

B. the strong electrostatic attraction between oppositely charged ions

C. the attraction between positive ions and a sea of delocalised electrons

D. the weak attraction between neighbouring molecules

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9 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

Calcium carbonate reacts with dilute hydrochloric acid.

(a) Give the word equation for this reaction. [2]

(b) Complete the balanced equation for this reaction, including the missing state symbols.

$CaCO_3(s) + ...HCl(aq) \rightarrow CaCl_2(....) + H_2O(....) + CO_2(....)$ [2]

(c) When aluminium powder is heated strongly with iron(III) oxide, Fe_2O_3, the products are aluminium oxide and iron. The iron forms as a liquid. All the other substances in the reaction are solids. Give a balanced equation for this reaction, including state symbols. [2]

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10 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

Ammonium nitrate, NH_4NO_3, and urea, $CO(NH_2)_2$, are both used as fertilisers. Use the Periodic Table.

(a) State the unit used for the amount of a substance. [1]

(b) Show that the relative formula mass (Mr) of ammonium nitrate is 80. [1]

(c) Calculate the amount, in moles, of ammonium nitrate in a 2.0 kg bag of ammonium nitrate. [2]

(d) Calculate the mass, in grams, of 0.40 mol of urea. [2]

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11 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

Sodium burns in chlorine gas to form sodium chloride. In an experiment, 4.6 g of sodium reacts completely with chlorine. Use the Periodic Table.

(a) Complete the balanced equation for this reaction.

....Na(s) + Cl_2(g) → ....NaCl(s) [1]

(b) Calculate the maximum mass of sodium chloride that can form. [3]

(c) Calculate the mass of chlorine that reacts with the 4.6 g of sodium. [2]

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12 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [7]

Calcium oxide is made in a lime kiln by heating calcium carbonate strongly.

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

In one batch, 50.0 kg of calcium carbonate is heated. Use the Periodic Table.

(a) Calculate the maximum mass, in kg, of calcium oxide that could form. [3]

(b) The actual mass of calcium oxide obtained was 25.2 kg. Calculate the percentage yield of calcium oxide. [2]

(c) Suggest two reasons why the percentage yield was less than 100%. [2]

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13 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [6]

Compound X contains only carbon, hydrogen and oxygen. By mass, it contains 54.5% carbon, 9.1% hydrogen and 36.4% oxygen. Use the Periodic Table.

(a) State what is meant by the term empirical formula. [1]

(b) Calculate the empirical formula of compound X. [3]

(c) The relative formula mass (Mr) of compound X is 88. Deduce the molecular formula of compound X. Show your working. [2]

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14 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [7]

Hydrated magnesium sulfate has the formula $MgSO_4 \cdot xH2O$. Leon heats a sample of hydrated magnesium sulfate in a crucible to remove the water of crystallisation. He heats, cools and weighs the crucible several times. The table shows his results.

Measurement	Mass in g
empty crucible	25.30
crucible + hydrated magnesium sulfate	30.22
crucible + contents after first heating	28.05
crucible + contents after second heating	27.72
crucible + contents after third heating	27.70
crucible + contents after fourth heating	27.70

(a) Explain why Leon continued heating until the mass stopped changing. [2]

(b) Calculate the mass of anhydrous magnesium sulfate left in the crucible and the mass of water removed. [2]

(c) Calculate the value of x in $MgSO_4 \cdot xH2O$. Use the Periodic Table. [3]

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15 · Part 2

Paper 1 & 2 | Chemical formulae, equations and calculations | [8]

Maryam wants to find the formula of magnesium oxide by heating magnesium ribbon in air. She has a crucible with a lid, a pipe-clay triangle, a tripod, a Bunsen burner and a balance.

(a) Describe how Maryam could use this apparatus to find the mass of magnesium used and the mass of magnesium oxide formed. [4]

(b) The table shows Maryam's results. Calculate the empirical formula of magnesium oxide. Use the Periodic Table.

Measurement	Mass in g
crucible + lid	22.48
crucible + lid + magnesium	22.84
crucible + lid + magnesium oxide	23.08

[3]

(c) Give a reason why the lid is lifted only slightly, and only from time to time, during heating. [1]

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16 · Part 2

Paper 2 only Chemical formulae, equations and calculations [6]

Tomasz dissolves 5.85 g of sodium chloride, NaCl, in water and makes the solution up to a total volume of 250 cm^3. Use the Periodic Table.

(a) Calculate the concentration, in mol/dm^3, of this sodium chloride solution. [3]

(b) Calculate the amount, in moles, of sodium chloride in 25.0 cm^3 of this solution. [1]

(c) Tomasz adds water to 25.0 cm^3 of the solution until the total volume is 200 cm^3. Calculate the concentration, in mol/dm^3, of the diluted solution. [2]

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17 · Part 2

Paper 2 only

Chemical formulae, equations and calculations [7]

In a titration, 25.0 cm^3 of sodium hydroxide solution is neutralised by sulfuric acid of concentration 0.100 mol/dm^3.

(a) Complete the balanced equation for this reaction.

$H_2SO_4(aq) + ...NaOH(aq) \rightarrow Na_2SO_4(aq) + ...H_2O(l)$ [1]

(b) The table shows the volumes of sulfuric acid used. Calculate the mean volume of sulfuric acid, ignoring the rough titration.

Titration	**rough**	**1**	**2**	**3**
Volume of sulfuric acid in cm^3	20.80	20.05	19.95	20.00

[1]

(c) Calculate the concentration, in mol/dm^3, of the sodium hydroxide solution. [4]

(d) Give a reason why the rough titration is not used when calculating the mean volume. [1]

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18 · Part 2

Paper 2 only

Chemical formulae, equations and calculations [7]

Hydrogen peroxide solution decomposes to form water and oxygen when a catalyst is added.

$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

Use the Periodic Table. (1 mol of any gas occupies 24 dm^3 at rtp)

(a) A solution contains 0.68 g of hydrogen peroxide, H_2O_2. Calculate the volume, in cm^3, of oxygen at rtp formed when all of this hydrogen peroxide decomposes. [3]

(b) Calculate the mass of 900 cm^3 of oxygen gas at rtp. [2]

(c) Explain why 100 cm^3 of carbon dioxide gas has a greater mass than 100 cm^3 of oxygen gas, when both are measured at rtp. [2]

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19 · Part 2

Paper 2 only Chemical formulae, equations and calculations [8]

Priya adds 1.95 g of zinc powder to 50.0 cm^3 of dilute sulfuric acid of concentration 0.500 mol/dm^3.

$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$

Use the Periodic Table. (1 mol of any gas occupies 24 dm^3 at rtp)

(a) Calculate the amount, in moles, of zinc added. [1]

(b) Calculate the amount, in moles, of sulfuric acid used. [1]

(c) Explain which reactant is in excess. [2]

(d) Calculate the volume, in cm^3, of hydrogen formed at rtp when the reaction stops. [2]

(e) Calculate the mass of zinc left unreacted when the reaction stops. [2]

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20 · Part 2

Paper 2 only

Chemical formulae, equations and calculations [7]

An oxide of copper is placed in a reduction tube and heated in a stream of hydrogen gas until only copper remains. The water vapour formed leaves the tube. Hydrogen is still passed through the tube while the copper cools. The table shows the results.

Measurement	Mass in g
empty reduction tube	30.15
tube + oxide of copper	33.33
tube + copper after the reaction	32.69

(a) Calculate the empirical formula of the oxide of copper. Use the Periodic Table. [3]

(b) Explain why hydrogen is still passed through the tube while the copper cools. [2]

(c) Each oxygen atom removed from the oxide combines with one molecule of hydrogen, H_2, to form water. Calculate the minimum volume, in cm^3, of hydrogen at rtp that reacts in this experiment. (1 mol of any gas occupies 24 dm^3 at rtp) [2]

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21 · Part 2

Paper 1 & 2 | Ionic bonding | [6]

Calcium reacts with chlorine to form calcium chloride, an ionic compound. The electronic configuration of a calcium atom is 2.8.8.2 and that of a chlorine atom is 2.8.7.

(a) State the charge on a calcium ion and the charge on a chloride ion. [2]

(b) Describe what happens to the electrons when calcium atoms react with chlorine atoms to form calcium chloride. [3]

(c) Give the formula of calcium chloride. [1]

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22 · Part 2

Paper 1 & 2 Ionic bonding [6]

(a) Complete the table by giving the formula of each compound.

Compound	Formula
potassium oxide	
iron(III) hydroxide	
ammonium carbonate	
zinc nitrate	

[4]

(b) Nitrogen is in Group 5 of the Periodic Table. Explain why the formula of calcium nitride is Ca_3N_2. [2]

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23 · Part 2

Paper 1 & 2 Ionic bonding [6]

Sodium chloride and magnesium oxide are both compounds with giant ionic lattices. Sodium chloride melts at 801 °C. Magnesium oxide melts at 2852 °C.

(a) Explain why sodium chloride has a high melting point. [4]

(b) Explain why magnesium oxide has a much higher melting point than sodium chloride. [2]

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24 · Part 2

Paper 1 & 2 Ionic bonding [6]

Zinc chloride is an ionic compound. Ali tests samples of zinc chloride using a circuit containing a battery, a lamp and two graphite electrodes. The table shows his results.

Sample of zinc chloride	Does the lamp light?
solid	no
molten	yes
dissolved in water	yes

(a) Explain these results. [4]

(b) Zinc chloride melts at 290 °C and sodium chloride melts at 801 °C. Suggest why zinc chloride, rather than sodium chloride, is used to test a molten ionic compound in a school laboratory. [1]

(c) Give the formula of zinc chloride. [1]

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25 · Part 2

Paper 1 & 2 Ionic bonding [7]

Rubidium is in Group 1 of the Periodic Table. Selenium is in Group 6. Rubidium reacts with selenium to form an ionic compound, rubidium selenide.

(a) Give the formulae of the two ions in rubidium selenide and deduce the formula of rubidium selenide. [2]

(b) The table shows some properties of three substances, J, K and L. Identify which substance is an ionic compound. Explain your answer using the data in the table.

Substance	Melting point in °C	Conducts electricity when solid?	Conducts electricity when molten?
J	1083	yes	yes
K	714	no	yes
L	114	no	no

[3]

(c) Explain, in terms of electrons, why a rubidium ion has a charge of 1+ and a selenide ion has a charge of 2−. [2]

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